Arithmetic Brownian Motion: Exact Solution
hasklib mathematical companion

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Abstract

This note derives the exact terminal law of arithmetic Brownian motion (ABM), the process underlying Shreve (2004)'s classical treatment and, historically, Bachelier's original 1900 model of a stock price. Unlike GBM and OU, neither Itô's lemma nor an integrating factor is needed: both coefficients are already constant, so the SDE integrates directly.

1 The ABM SDE

Definition 1 (Arithmetic Brownian motion)**.** X_t follows arithmetic Brownian motion with drift $\mu \in \mathbb{R}$ and volatility $\sigma > 0$ if it solves

$$\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad X_0 \in \mathbb{R}. \tag{1}$$

In the notation of `ito_foundations.tex` this is $a(t,x) = \mu$, $b(t,x) = \sigma$ — both constant, with no x-dependence at all.

Remark 1 (Correspondence with the code). `ABM::drift` returns the constant μ and `ABM::diffusion` returns the constant σ, independent of x in both cases. Since neither coefficient depends on x, the Lipschitz and linear-growth conditions of the existence–uniqueness assumption in `ito_foundations.tex` hold trivially — this is the simplest possible case of that assumption.

2 Exact solution

Both GBM and OU needed a substitution to remove an X_t-dependence from the coefficients before the SDE could be integrated: $\ln x$ for GBM, the integrating factor $e^{\kappa t}$ for OU. Here there is nothing to remove — μ and σ are already constants — so the general relation from `ito_foundations.tex`,

$$X_t = X_0 + \int_0^t a(s, X_s)\,\mathrm{d}s + \int_0^t b(s, X_s)\,\mathrm{d}W_s,$$

integrates immediately once $a = \mu$ and $b = \sigma$ are substituted.

Theorem 1 (Exact terminal law of ABM)**.** *Let X_t solve* (1)*. Then for any $T > 0$,*

$$X_T = X_0 + \mu T + \sigma W_T, \tag{2}$$

where $W_T \sim \mathcal{N}(0, T)$, so

$$X_T \sim \mathcal{N}\big(X_0 + \mu T,\ \sigma^2 T\big). \tag{3}$$

Proof. Substituting $a(s, X_s) = \mu$ and $b(s, X_s) = \sigma$,

$$X_T = X_0 + \int_0^T \mu \, \mathrm{d}s + \int_0^T \sigma \, \mathrm{d}W_s = X_0 + \mu T + \sigma \int_0^T \mathrm{d}W_s.$$

The remaining integral is $\int_0^T \mathrm{d}W_s = W_T - W_0 = W_T$, giving (2). Since $X_0 + \mu T$ is a fixed number and $W_T \sim \mathcal{N}(0, T)$, the affine transformation $X_0 + \mu T + \sigma W_T$ is itself Gaussian with mean $X_0 + \mu T$ and variance $\sigma^2 \operatorname{Var}(W_T) = \sigma^2 T$, giving (3). □

Remark 2. Equation (3) is sampled with a single normal draw: generate $Z \sim \mathcal{N}(0, 1)$ and set $X_T = X_0 + \mu T + \sigma\sqrt{T}\, Z$. As with `GBM::sample_terminal` and `OU::sample_terminal`, no time-stepping is needed — this is what `ABM::sample_terminal` implements.

3 Moments and the relation to OU

Theorem 1 already gives the moments directly, with no separate lemma needed:

$$\mathbb{E}[X_T] = X_0 + \mu T, \qquad \operatorname{Var}(X_T) = \sigma^2 T.$$

Unlike OU, the variance grows without bound as $T \to \infty$: there is no mean-reverting pull holding the process in, so uncertainty simply accumulates. Correspondingly, $\mathbb{E}[X_T]$ grows in a straight line rather than curving toward a long-run level — there is no θ here for it to curve toward.

Remark 3 (ABM as a degenerate OU process). Setting $\kappa = 0$ in the OU drift $\kappa(\theta - X_t)$ removes it entirely, leaving $\mathrm{d}X_t = \sigma \, \mathrm{d}W_t$ — driftless Brownian motion. ABM generalises this by adding back a constant drift μ that is *not* tied to any mean-reversion mechanism. Equivalently: ABM is what OU degenerates into once mean-reversion is switched off, plus an unconstrained drift term OU's $\kappa = 0$ case does not have on its own.

References

Shreve, S. E. (2004), *Stochastic Calculus for Finance II: Continuous-Time Models*, Springer Finance, Springer, New York.